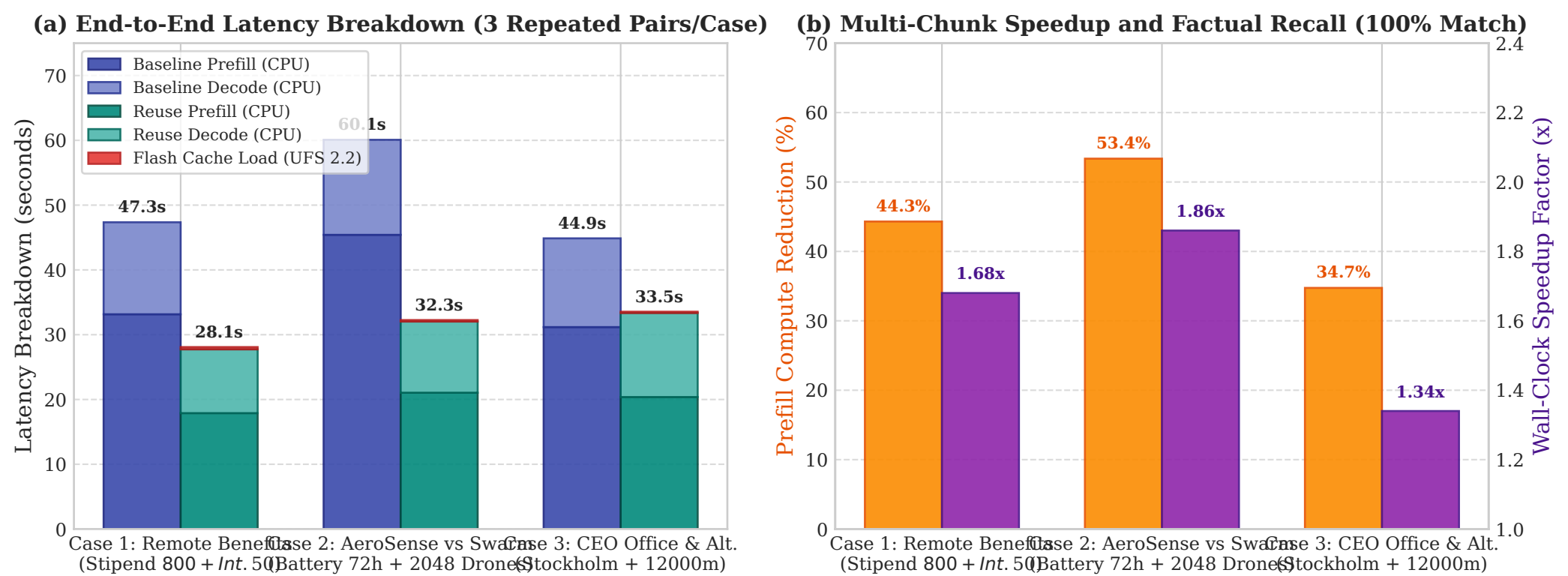
(a) End-to-End Latency Breakdown (3 Repeated Pairs/Case)
Baseline Prefill (CPU)
Baseline Decode (CPU)
Reuse Prefill (CPU)
Reuse Decode (CPU)
Flash Cache Load (UFS 2.2)
Latency Breakdown (seconds)
47.3s
28.1s
60.1s
32.3s
44.9s
33.5s
Case 1: Remote Benefits (Stipend 800 + Int. 50)
Case 2: AeroSense vs Swarm (Battery 72h + 2048 Drones)
Case 3: CEO Office & Alt. (Stockholm + 12000m)
(b) Multi-Chunk Speedup and Factual Recall (100% Match)
Prefill Compute Reduction (%)
Wall-Clock Speedup Factor (x)
44.3%
1.68x
53.4%
1.86x
34.7%
1.34x